



Personality, grit, and performance in rock-climbing: down to the nitty-gritty

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ABSTRACT

Over the past decades, the study of personality gained momentum in the field of sport psychology, a prime example being the increased number of publications relating personality to performance in extreme sports. The aim of the present investigation was twofold: to relate Five-Factor Model (FFM) personality traits with various aspects of climbing performance and to estimate whether grit, as a distinct personality trait, predicts climbing performance over and beyond FFM dimensions. Our sample included adult bouldering and sport climbing practitioners ($N = 272$ sport climbers, 155 boulderers) with ages between 16 and 69 ($M = 32.1$, $SD = 10.0$). We measured personality (Big Five Inventory FFM–2 Short Form, and a 12-item Grit Scale) and multiple indicators of outdoor sport climbing and bouldering performance. Results indicated that climbing performance was predicted by openness and agreeableness. Grit also significantly predicted climbing performance over the FFM traits. Contrary to the commonly held view depicting grit as indistinct from conscientiousness, our findings suggest that this trait has a unique contribution to explaining performance in a relatively novel, high-risk sport.

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Research on personality and its implications for various health, work-related or athletic outcomes remained popular throughout the decades. Its importance with respect to athletic performance or discrete athletic behaviours elicited the interest of many sport psychologists, practitioners, and researchers alike (e.g., Laborde et al., 2019; Roberts et al., 2018; Waleriańczyk & Stolarski, 2021). The influential review authored by Allen et al. (2013) concluded that we already have substantial evidence to indicate that personality predicts long term athletic success, as well as various short-term sport-relevant behaviours. Moreover, personality coherently differentiates between professional athletes and non-athletes across a range of disciplines (Allen et al., 2013). Extant research suggests that various traits have significant associations with a range of performance indicators or discrete athletic behaviours across several athletic disciplines. Professional athletes who

regularly train and attend competitions exhibit higher levels of extraversion and conscientiousness, and lower levels of neuroticism compared to people regularly practicing physical activities, but not on a professional level (Allen et al., 2013). Athletic performance across various sports appears to be related mostly with extraversion, neuroticism (emotional stability) and conscientiousness. One factor undermining the strength of the conclusions emerging from various reviews is the relatively small number of empirical articles investigating these associations. A larger number of empirical investigations would allow a more accurate estimation of the role of personality in sport-related outcomes (Allen et al., 2013). We aim to contribute to the general literature on personality in sport by investigating its role in a relatively novel and under-researched discipline: rock-climbing.

Personality in sports. Although various personality taxonomies have been used to understand how it relates to athletic performance, the Five Factor Model (FFM) emerged as the dominant framework (e.g., Allen et al., 2013). A growing body of research identified significant associations between FFM dimensions (neuroticism, extraversion, openness, agreeableness, and conscientiousness) and performance in various sports. For example, Allen et al. (2015) and Möttus et al. (2017) reported that extraversion and conscientiousness were significantly associated with the time spent in physical activity. FFM traits were found to have significant and direct associations with various facets of athletic performance. For example, Steca et al. (2018) showed that athletes experiencing success in their discipline scored higher than non-athletes in emotional stability, agreeableness, conscientiousness, and extraversion. High-level athletes competing at international levels have been shown to be more conscientious and emotionally stable compared to athletes participating in local competitions (Allen et al., 2011). Martin et al. (2011) showed that athletes who qualified for Paralympic games had lower levels of agreeableness (higher tough-mindedness facet scores) and neuroticism (lower anxiety facet scores). The longitudinal associations between personality and progression to professional sport hover around the same personality traits, adolescent practitioners that became professional athletes having higher levels of conscientiousness and lower levels of neuroticism, respectively (Aidman, 2007).

Personality traits have been associated not only with broad sport-related outcomes, but also with discrete athletic behaviours that facilitate attaining performance in the respective disciplines. For example, conscientiousness predicted the use of better preparation strategies (Woodman et al., 2010) and more effective coping strategies before and during competitive events (Kaiseler et al., 2012). Kaiseler et al. (2019) identified that all the FFM traits were significantly associated with various forms of sport-related coping among athletes. Neuroticism was associated with positive cognitive and emotional responses to unsuccessful outcomes (Allen et al., 2014).

We pursue two overarching objectives: first, investigating whether the aforementioned associations generalise to relatively new sports, sports labelled as “high-risk” or “extreme”, in our case, rock climbing; second, we seek to identify whether additional personality-related variance in rock-climbing performance can be uncovered by considering the role of grit, a personality trait placed outside the FFM.

Personality in high-risk sports. During the past decades, high-risk sports have increased in popularity, attracting a growing number of practitioners (Clough et al., 2016), culminating in 2020 with the inclusion of Sport Climbing or Surfing on the list of Olympic

Disciplines introduced in the 2021 Tokyo and 2024 Paris Olympic Games. High-risk sports were defined as disciplines where mismanaged mistakes or accidents can lead to severe or potentially fatal injuries (Brymer et al., 2020). To date, little is known about how psychological factors come into play in such sports. Practicing these sports requires an array of psychological functions such as cognitive dexterity, courage, and the ability to act in environments that pose a risk to a person's life (McEwan et al., 2019). We seek to identify whether personality traits impact performance in rock-climbing.

FFM traits have been shown to be relevant for participation in high-risk sports such as surfing (Diehm & Armatas, 2004), scuba diving, free diving, paragliding, rafting, and rock-climbing (Tok, 2011). McEwan et al.'s (2019) review reported that Extraversion had a significant association with participation in extreme sports. Tok (2011) also reported that sensation seeking and extraversion were positively related to high-risk sport participation, while neuroticism was negatively related to the same outcome. Considering that managing fear, anxiety, and specific phobias (fears of heights) is typically part of the "mental fabric" of high-risk sports practitioners, the aforementioned associations are not surprising. In comparison with athletes practicing low-risk sports, the practitioners of high-risk sports had higher levels of extraversion and openness, while having lower levels of conscientiousness and neuroticism (e.g., Allen et al., 2011; Tok, 2011). As for the remaining FFM traits, no previous associations with practicing extreme sports have been identified in the meta-analytical investigation conducted by McEwan et al. (2019).

We were able to identify a single investigation focusing specifically on personality and participation in rock-climbing (Rumbold et al., 2021). In this trail-blazing empirical investigation, personality profiles, rather than individual traits were analysed in a sample of rock-climbing practitioners. The findings reported by Rumbold et al. (2021) challenged the conventional views that practitioners of such sports are homogeneously "sensation-seekers" by identifying four distinct personality profiles: Healthy (higher scores on openness and lower scores on neuroticism), Emotionally Unstable (higher scores on neuroticism), Measured and Compliant (higher observed scores on agreeableness and conscientiousness), and Curious and Impulsive (higher scores on openness). Additionally, they explored how the profiles differed in respect to sensation-seeking. Out of all the FFM dimensions, openness was the "most salient big five factor that determines high levels of sensation-seeking in climbers" (Rumbold et al., 2021, p. 5). While informative in respect to distinguishing typical behaviours and preferences in rock-climbing, and personality's role in sensation-seeking, this study does not directly address the impact of personality on rock-climbing performance, as data on sport climbing or bouldering performance was not collected. Moreover, Rumbold et al. (2021) sampled practitioners from different forms of climbing (sport climbing, bouldering and trad climbing), not specifically differentiating between the disciplines.

To summarise, the current understanding regarding the role of personality in high-risk sports is limited by several factors, such as (1) extremely small sample sizes – the 39 empirical investigations included in McEwan et al.'s (2019) review had a median sample size of 21 participants, with samples ranging from 2 to 255 participants; resulting in (2) insufficient power (considering that personality had small to mid-sized effects on sport participation and/or performance, such sample sizes are insufficient for detecting such effects); (3) difficulty in assessing participants within a specific discipline, as most of the existing investigations relied on multi-sports (rock-climbing, snowboarding and

windsurfing), such sports rarely requiring the same array of physical or mental characteristics (e.g., Dudek et al., 2016); (4) difficulty in estimating athletic performance within disciplines labelled as high-risk sports, as most of them do not operate within a standardised framework for accurately operationalising performance; (5) sample composition – males to females' ratio in the empirical studies included in McEwan et al.'s (2019) review was almost 3–1.

Current investigation

The overarching objective of our investigation is to expand the current scientific understanding regarding the relationship between personality and performance in sport climbing and bouldering.

Our investigation focuses exclusively on two different forms of rock-climbing, which brings focus and precision to the target group of the research, resulting in several advantages. First, we draw from a sufficiently large sample of sport climber and boulderers, mitigating potential biases stemming from aggregating different athletic disciplines. Second, we investigate how personality traits relate to different performance criteria across the two disciplines. Third, we investigate whether grit, a relatively novel trait is related with sport performance criteria over and beyond FFM traits. Our first objective is to expand the current scientific understanding regarding the role of personality in attaining performance in high-risk sports by investigating whether the FFM traits predict performance in rock-climbing. Consequently, we hypothesise that:

H1: FFM personality traits will account for a significant proportion of variance in rock-climbing performance, when controlling for age, gender, and years of experience in rock-climbing.

In line with the extant research, we expect that not all FFM traits will have an equal contribution to explaining differences in rock-climbing performance. Building on the previous findings concluding that openness (e.g., Rumbold et al., 2021), extraversion (e.g., Wilson & Dishman, 2015) and neuroticism (e.g., McEwan et al., 2019) consistently predict high-risk sport participation, we expect these traits to account for most of the personality-related variance in rock-climbing performance. Consequently, we hypothesise that:

H1a: Most of the personality-related variance in rock-climbing performance will be explained by neuroticism, extraversion, and openness.

The current investigation responds to the call for building a more substantial body of empirical research about personality and sport and athletic performance (Allen et al., 2013). Apart from the climbing performance conclusions that might be appealing only to professionals and athletes practicing the sport, understanding the role of personality in various forms of athletic performance, especially in high-risk contexts can be insightful for other areas of applied psychology and human performance where similar high-risk levels are present.

Grit: a predictor of long-term success. Despite the major developments obtained by using a single framework for capturing personality preferences, namely the FFM, several authors suggested that research should move beyond this model for grasping the impact of personality in sport (Laborde et al., 2019). One trait sitting outside the

FFM that was associated with success in long-term endeavours is grit. Grit was defined as “perseverance and passion for long term goals. Grit entails working strenuously toward challenges, maintaining effort and interest over years despite failure, adversity, and plateaus in progress” (Duckworth et al., 2007, p. 1087). Grit has been associated with important life outcomes, explaining a unique, albeit minor proportion of variance in academic success or job retention, beyond other personality dimensions (Duckworth et al., 2007; Eskreis-Winkler et al., 2014).

Extant research suggests that grit was associated with various adaptive behaviours and sport-related outcomes. For instance, grit had a significant association with athletic identity (Mosewich et al., 2021). Grit moderated the relationship between motivational feedback and subsequent athletic performance (Moles et al., 2017). Moreover, grit emerged as a significant predictor of sport-specific activities, including training, play, or participating in competitions (Larkin et al., 2016). Toering and Jordet (2015) demonstrated that grit had a positive association with self-restraint and impulse control in elite soccer players. Apart from its direct relationship with athletic practice and performance, recent research reported a significant association between grit and different mental aspects potentially relevant for rock-climbing practice. For example, an empirical investigation revealed that grit was related with the experience of flow, deep and effortless concentration (Smith et al., 2020). More recent empirical investigations illustrate that grit explains deliberate practice and less ideation on quitting the sport (Tedesqui & Young, 2018). Grit, in the form of passion, was found to be relevant for expert performance, having the potential to distinguish elite athletes from non-athletes (e.g., From et al., 2020; Newland et al., 2020). Some drew on Bandura’s (2001) Social Cognitive Theory for explaining grit’s role in athletic performance (e.g., Newland et al., 2020), arguing that grit is a reflection of Bandura’s intentionality (purposeful intention towards action) and perseverance through adversity in attaining higher-level goals.

The passion and perseverance in attaining long-term objectives might be particularly relevant to rock-climbing performance. Popular accounts about elite-level athletes such as Chris Sharma, Adam Ondra or Nalle Hukkataival spending months or even years to successfully complete a route are not uncommon inside climbing community. Additionally, the necessity of constant, uninterrupted climbing-specific training was emphasised in various reviews or empirical studies. For example, Saul et al. (2019) concluded that uninterrupted exposure to climbing-specific training represented “... the best way to succeed ... and push the redpoint grade” (Saul et al., 2019, p. 97). This idea was supported also by empirical research outlining the importance of constant and dedicated training for acquiring the major physical and physiological determinants of climbing performance, such as hand and finger-strength, shoulder endurance, power, or power-endurance (MacKenzie et al., 2020). Consequently, being regarded as a trait that drives the individual towards attaining longer term objectives via persistence and dedication, grit could be one of the key psychological determinants of rock-climbing performance.

However, despite such encouraging previous findings, grit-related research was also plagued by a major limiting factor: grit’s insufficient differentiation from conscientiousness. The two constructs seemed to be difficult to distinguish, presenting “phenotypic correlations of approximately .70” (Rimfeld et al., 2016, p. 2). This lack of differentiation from conscientiousness was further tackled by (Credé et al., 2017) in a meta-analytical investigation. Three major conclusions surrounding grit emerged in the afore-mentioned

meta-analysis: first, grit did not emerge as a higher-order personality trait, having two distinct lower-order facets, perseverance of effort and consistency of interests; second, grit exhibited significant, albeit modest relations with various criteria, such as academic performance or retention; third, due to its overlap with conscientiousness, Credé et al. (2017) argued that grit's incremental validity in respect to various criteria could be limited. Despite these limitations, Credé et al. (2017) reported that perseverance of effort had significant criterion-related validity in respect to academic performance, even when controlling for conscientiousness. To conclude, although grit might be appealing from an intervention standpoint, any empirical grit-oriented investigation should adequately control for the potential overlaps with conscientiousness.

Consequently, encouraged by the previous findings, but also mindful of the potential pitfall – the common variance between grit and conscientiousness, our second overarching objective is to test whether grit predicts performance in rock-climbing, over and beyond FFM personality traits, particularly over conscientiousness.

H2: Grit will account for a significant proportion of variance in rock-climbing performance, over age, gender, experience and FFM traits, in particular over and beyond conscientiousness.

Methods

Participants

Power analysis. To determine the sample size needed for detecting small-to-medium effect sizes, we conducted an *a-priori* power analysis, by using the *G-Power* software, version 3.1, following the approach outlined by Faul et al. (2007). Our analysis indicated that the total number of participants needed for detecting such effects ($f^2 > .10$), for a linear multiple regression that includes nine predictors, with a power of .80, was 159 participants. In respect to the sample size needed for conducting confirmatory factorial analyses, we used the participants (*N*) to number of variables (*p*) ratio. The recommended *N:p* ratios range between a minimum of 5 and an optimal ratio of 10 (e.g., Wang & Wang, 2012). Our measures included 30 (the personality measure), 12 (grit measure) and 4 observed indicators (performance in rock-climbing). The smallest *N:p* ratio was 9.06, for the personality measure. The remaining ratios were situated well above the optimal recommended 10:1 threshold.

Following institutional ethics approval, we collected data from 272 participants (155 boulderers) who volunteered to respond to our invitation (see Procedure). The sample included 114 females (41.91%) and 158 males, with ages between 16 and 69 ($M = 32.13$, $SD = 10.01$) – with parental consent for participants under 18 years old. To be included in the current study the participants had to have at least 12 months of experience in practicing rock-climbing. In respect to nationality, 80 participants (29.4%) were from US and Canada, 63 participants (23.2%) from the UK, 107 participants (39.3%) from various EU countries (Austria, Belgium, France, Germany, Italy, Romania, Spain) and the remainder of 22 participants (8.1%) were from various countries (Australia, Brazil, China, Japan, and New Zealand, South Africa, Thailand). Their experience in rock-climbing ranged between 1 and 40 years ($M = 9.02$, $SD = 9.14$). In respect to education, 107 participants (39.3%) attained a graduate or post-graduate degree, 120 participants (44.1%) attained a bachelor's degree, 107

participants had a graduate or post-graduate degree, and 45 participants (16.5%) had a high school degree or equivalent.

Measures

Climbing Performance. Different forms of rock-climbing can be distinguished, depending several criteria on the type of protection that the climbers use, type of terrain that is being climbed, length of the routes, climbing styles: free soloing (no protection used), aid climbing (using artificial equipment to move up the wall), bouldering (using crashpads as protection, typically not higher than 5–7 m), sport climbing (climbers are expected to progress on a route that includes fixed anchors, being protected by the rope that is passed through each anchor, but without using it to progress up the route), trad climbing (the same approach as sport climbing, but using mobile anchors such as friends or nuts instead of fixed anchors) and many other forms. However, based on their popularity and on the possibility to organise competitions, two forms of rock-climbing have been included on the list of Olympic athletic disciplines: sport climbing and bouldering, therefore they constituted the object of interest for our study. In estimating performance within these two disciplines, two components are taken into account: route difficulty level and climbing style. In respect to difficulty, different scales have been proposed by various national or international mountaineering and/or climbing associations: the French/sport scale, the Yosemite Decimal System (YDS), the Union Internationale des Associations d'Alpinisme scale (UIAA), and the Ewbank scale. To complicate matters even further, difficulty ratings in bouldering were reported by using either the Font or Vermin scales. To bring a greater degree of uniformity in reporting performance in various rock-climbing disciplines, the International Rock-Climbing Research Association (IRCRA) issued a position statement outlining a unified scale that permits the conversion of various grading systems into a single rock-climbing difficulty scale (IRCRA; Draper et al., 2015). The IRCRA difficulty scale was included in Appendix 1.

As for the element of style, when measuring performance within the two disciplines, athletes report not only the highest route difficulty they have completed, but also the “style” in which it was completed: redpoint, flash and onsight. Redpoint (original *Rot-Punkt*) refers successfully climbing a route (with rope), without any artificial aid, after having had the opportunity to experience the respective route for as long as the climber finds it necessary. The process via which the climber becomes familiarised with the route is referred to as projecting. When athletes successfully completed extremely hard routes, especially for grades that were not completed until that point, the process of projecting can last months or even years. Flash refers to successfully ascending the route (with rope) without having actually had any physical contact with the route but having the opportunity to learn about the route's specific demands by viewing it (either via videos or via oral accounts from persons who completed the route). Onsite refers to the style of climbing a route (with rope) without any artificial aid and without having any kind of information about the route (no previous visual analysis via pictures or videos and no other account about the route's characteristics).

Drawing from an international sample of rock-climbing participants, each participant was allowed to select the most familiar reporting scale (French, UIAA, YDS or Ewbank). The reported grades have been converted according to the recommendations outlined

in IRCRA's position statement (IRCRA; Draper et al., 2015). The participants reported the most difficult routes they managed to successfully complete in the three different styles outlined above. Because bouldering involves a smaller number of moves and the athlete can visually identify most of the routes' characteristics, the conditions for "onsighting" are impossible to meet, therefore the highest flash is typically reported.

Personality. In order to ensure a brief, but accurate estimation of the FFM traits, personality was measured via Big Five Inventory–2 Short Form (BFI-2-S; Soto & John, 2017). The five domains were measured with 30 items rated on a 5-point scale from 1 (*strongly disagree*) to 5 (*strongly agree*). Internal consistencies for the five dimensions ranged between .72 (Conscientiousness) and .82 (Neuroticism).

Grit. Grit was measured by employing the 12-item inventory (Duckworth et al., 2007). Some example items are "Setbacks don't discourage me" or "I often set a goal but later choose to pursue a different one", measured on a scale from 1 (*not like me at all*) to 5 (*very much like me*). Internal consistency reliability as measured with Cronbach's Alpha was .79.

Procedure

The measures were administered via an online survey. Considering the challenges in identifying and enrolling a sufficiently large number of practitioners, we decided to open the survey to the international community of rock climbing and bouldering practitioners. Consequently, the survey was disseminated via email and social networks to English-speaking adult climbers from different countries. The participants were required to be fluent in English to complete the survey. The participants were explained their rights as volunteers and all of them provided their informed consent before completing the survey. No incentives were offered.

Data analysis

Following the recommendations of Ployhart et al. (2003), we conducted confirmatory factor analyses (CFA) for all variables (Appendix 2, Appendix 3). Personality was the only exception, due to concerns regarding the use of CFA for personality. Several straightforward and seemingly valid personality frameworks were not confirmed via CFA, (e.g., Borkenau & Ostendorf, 1990; Church & Burke, 1994; Parker et al., 1993), leading some authors to raise concerns about the utility of this technique for analysing personality data. Most of the critiques have focused on the imposition of zero order constraints on covariances outside of the perfect predicted factor structure. Imposing this constraint was deemed inappropriate for analysing personality structures, as they are characterised by complex item-construct inter-relations (e.g., Aluja et al., 2005; Marsh & Hau, 2007). Consequently, we conducted exploratory structural equation modelling (ESEM) for the personality measure (Appendix 4). ESEM employs the traditional CFA parameters as fit indices, residual correlations or factor regressions, while deploying a rotation of the initial measurement model. This way ESEM permits the testing of a theoretical model without a strict specification of the measurement structure (Asparouhov & Muthén, 2009). This technique has been successfully used in analysing personality data (e.g., Marsh et al., 2013).

Results

All the statistical analyses were performed with MPlus version 8.2 (Muthén & Muthén, 1998–2017). Means, standard deviations, bivariate correlations between our main variables and internal consistency reliabilities (Cronbach's Alpha) are shown in Table 1. Significant and mid-sized correlations between experience and most of the dimensions of climbing performance were observed. Generally, the FFM personality traits had non-significant relationships with the various dimensions of sport climbing performance, with a few exceptions. Neuroticism had a negative and significant correlation with two bouldering performance dimensions: highest redpoint over the last 12 months ($r = -.29$, $p < .001$) and highest flash ($r = -.17$, $p < .01$). The remaining dimensions had very weak and non-significant associations with the various dimensions of sport climbing performance.

The results for the CFA and ESEM analyses are reported in Table 2. As recommended by IRCRA (Draper et al., 2015), we fitted all the seven different dimensions of sport climbing (highest redpoint, highest redpoint during the past 12 months, highest three redpoints and highest onsight/flash) and bouldering performance, respectively (highest redpoint, highest redpoint during the past 12 months, highest three redpoints and highest flash) onto a two different latent factors (overall climbing performance and overall bouldering performance). The goodness of fit indices for this two-factor model of climbing performance was acceptable ($CFI = .943$, $RMSEA = .052$). The ESEM analysis deployed for short-form measure of the FFM allowed the items to freely load onto any of the five dimensions and exhibited acceptable goodness of fit ($CFI = .903$, $RMSEA = .051$). For the grit scale, the goodness of fit was acceptable ($CFI = .929$, $RMSEA = .058$). The initial model was amended by adding three correlated errors.

Next, we examined the criterion-related validity of the CFA-derived latent factors for FFM traits and grit over age and gender in predicting various facets of sport climbing performance and bouldering performance, respectively. To this end, we employed a hierarchical regression. Following recommendations in the literature (Hunsley & Meyer, 2003), age, gender and experience were entered in step 1. In Step 2 the FFM personality dimensions were included. In Step 3, grit was added to estimate its validity over both demographics and FFM traits. The results were reported in Tables 3 and 4. Four dimensions of climbing performance were considered: overall climbing performance (latent factors derived via CFA), highest redpoint, and highest onsight and flash, respectively.

Outdoor Sport Climbing Performance. The FFM dimensions that significantly predicted overall sport climbing performance (latent) were openness ($\beta = 3.21$, $p < .001$) and agreeableness ($\beta = -1.85$, $p < .001$). This accounted for a mere 7.4% of the overall variance in overall sport climbing performance over age, gender, and experience ($\Delta R^2 = .074$, $p < .001$). The FFM dimensions that significantly predicted highest redpoint were openness ($\beta = 3.29$, $p < .001$) and agreeableness ($\beta = -2.08$, $p < .001$). This accounted for a mere 6.5% of the overall variance in highest redpoint over age, gender, and experience ($\Delta R^2 = .065$, $p < .001$). Finally, the FFM dimensions that significantly predicted highest onsight were openness ($\beta = 2.03$, $p < .01$) and agreeableness ($\beta = -1.79$, $p < .01$). This accounted for a mere 6.7% of the overall variance in highest onsight over age, gender, and experience ($\Delta R^2 = .067$, $p < .001$). When included in step 3, the FFM dimensions that significantly predicted overall sport climbing performance were openness ($\beta = 2.89$, $p < .001$) and

Table 1. Descriptive statistics and intercorrelations.

	<i>M</i>	<i>SD</i>	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1 Age	32.1	10.0	–													
2 Experience	9.0	9.1	.69***	–												
3 Overall Sport climbing	16.0	4.5	.15*	.46***	–											
4 Highest Redpoint Sport climbing	17.3	5.0	.18**	.48***	.98***	–										
5 Highest Onsite Sport climbing	15.0	3.7	.13	.45***	.96***	.92***	–									
6 Overall Bouldering	10.9	9.9	–.03	.14*	.34***	.34***	.35***	–								
7 Highest Redpoint Bouldering	11.5	10.1	–.03	.14*	.35***	.35***	.36***	.99***	–							
8 Highest Flash Bouldering	18.1	3.7	.21**	.44***	.58***	.53***	.62***	.96***	.90***	–						
9 Neuroticism	15.5	5.1	–.13*	–.13*	–.06	–.05	.01	–.04	–.03	–.17*	(.82)					
10 Extraversion	20.5	4.4	–.05	–.03	.06	.04	.04	.02	.02	–.03	–.25***	(.74)				
11 Openness	23.0	3.9	.17**	.08	.09	.10	.07	–.03	–.04	.09	–.07	.21***	(.77)			
12 Agreeableness	22.3	4.3	.04	.00	–.09	–.08	–.12	–.09	–.09	.03	–.15	.04	.21***	(.74)		
13 Conscientiousness	21.1	3.9	.21**	.13*	.11	.12	.09	–.04	–.05	.13	–.27***	.04	.11	.27***	(.72)	
14 Grit	3.4	0.6	.23**	.20**	.21**	.22**	.17*	.09	.08	.24*	–.30***	.25**	.24**	.22**	.50***	(.79)

Note: *N* = 272.**p* < .05.***p* < .01.****p* < .001.

Table 2. Confirmatory and exploratory structural equations modelling.

Measure	Framework	Model	$\chi^2(df)$	CFI	RMSEA (90% CI)	SRMR
Climbing Performance	CFA	2 factor model	418.91(77)	.943	.052 (.038–.064)	.061
FFM-SF	ESEM	5 factors (5 correlated errors)	523.61 (290)	.903	.051 (.045–.057)	.044
Grit	CFA	1 factor (3 correlated errors) ^a	97.88 (51)	.929	.058 (.040–.075)	.055

^aNumber of correlated errors specified within each mode.

agreeableness ($\beta = -2.08, p < .001$), as well as grit ($\beta = 3.59, p < .001$). This accounted for a mere 2.7% of the overall variance in sport climbing performance over age, gender, experience, and FFM dimensions ($\Delta R^2 = .027, p < .001$). Furthermore, in step 3, the FFM dimensions that significantly predicted highest redpoint were openness ($\beta = 3.11, p < .001$) and agreeableness ($\beta = -2.37, p < .001$), as well as grit ($\beta = 3.81, p < .001$). This accounted for a mere 3.1% of the overall variance in the highest redpoint over age, gender, experience, and FFM dimensions ($\Delta R^2 = .031, p < .001$). Finally, in step 3, the FFM dimensions that significantly predicted the highest onsight were openness ($\beta = 1.95, p < .001$) and agreeableness ($\beta = -2.01, p < .001$), as well as grit ($\beta = 2.76, p < .01$). This accounted for a mere 2.9% of the overall variance in highest onsight over age, gender, experience, and FFM dimensions ($\Delta R^2 = .029, p < .01$).

Outdoor Bouldering Performance. The FFM dimensions that significantly predicted overall bouldering performance (latent) were openness ($\beta = 1.60, p < .01$) and agreeableness ($\beta = -1.28, p < .001$). This accounted for a mere 5.5% of the overall variance in overall

Table 3. Grit's incremental validity over age, gender, experience, and five factor personality traits in predicting outdoor sport climbing performance.

Step	Independent variable	Overall Sport Climbing Performance (latent)			Highest Redpoint			Highest Onsight		
		β	R^2	ΔR^2	β	R^2	ΔR^2	β	R^2	ΔR^2
1	Age	-.10	.221***	–	-.15*	.287***	–	-.13*	.271***	–
	Gender	-.86			-.81*			-.62*		
	Experience	.28***			.37***			.27***		
2	Age	-.12	.295***	.074***	-.17*	.351***	.065***	-.15	.337***	.067***
	Gender	-.55			-.51			-.37		
	Experience	.28***			.37***			.28***		
	Neuroticism	.43			.50			.45		
	Extraversion	.11			-.29			-.16		
	Openness	3.21***			3.29***			2.03**		
	Agreeableness	-1.85***			-2.08***			-1.79**		
	Conscientiousness	.40			.74*			.66		
3	Age	-.12	.322***	.027***	-.18	.382***	.031***	-.15	.366***	.029**
	Gender	-.66			-.65			-.45		
	Experience	.26			.35			.27		
	Neuroticism	.51			.62			.56		
	Extraversion	-.20			-.61			-.42		
	Openness	2.89***			3.11***			1.95***		
	Agreeableness	-2.08***			-2.37***			-2.01***		
	Conscientiousness	-.05			.26			.31		
	Grit	3.59***			3.81***			2.76**		

Note: β = Standardised β .

* $p < .05$.

** $p < .01$.

*** $p < .001$.

Table 4. Grit's incremental validity over age, gender, experience, and five factor personality traits in predicting outdoor bouldering performance.

Step	Independent variable	Overall Bouldering Performance (latent)			Highest Redpoint			Highest Flash		
		β	Adj. R^2	ΔR^2	β	Adj. R^2	ΔR^2	β	Adj. R^2	ΔR^2
1	Age	-.07	.212***	–	-.06	.142***	–	-.09	.247***	–
	Gender	-1.09*			-1.41*			-1.14*		
	Experience	.21***			.18**			.24***		
2	Age	-.09	.267***	.055**	-.05	.198***	.056	-.12	.278***	.032
	Gender	-.84*			-.63			-1.28		
	Experience	.20***			.16*			.24		
	Neuroticism	-.13			-1.46			-.23		
	Extraversion	-.79			-.72			-.67		
	Openness	1.60**			.11			1.41		
	Agreeableness	-1.28***			-.51			-.02		
	Conscientiousness	.67*			-.24			.37		
	Grit	2.60***			3.15*			2.97*		
3	Age	-.09	.287***	.020**	-.05	.221	.023*	-.12	.305***	.027*
	Gender	-.93			-.78			-1.39		
	Experience	.19			.15			.23*		
	Neuroticism	-.07			-1.44			-.20		
	Extraversion	-1.01			-.87			-.81		
	Openness	1.37**			-.48			.87		
	Agreeableness	-1.46***			-.69			-.18		
	Conscientiousness	.33			-.60			.05		
	Grit	2.60***			3.15*			2.97*		

Note: β = Unstandardised β .

* $p < .05$.

** $p < .01$.

*** $p < .001$.

bouldering performance over age, gender, and experience ($\Delta R^2 = .055$, $p < .01$). No other FFM dimensions significantly predicted this criterion. When included in step 3, the FFM dimensions that significantly predicted overall bouldering performance were openness ($\beta = 1.37$, $p < .01$) and agreeableness ($\beta = -1.46$, $p < .001$), as well as grit ($\beta = 2.60$, $p < .001$). This accounted for a mere 2% of the overall variance in overall bouldering performance over age, gender, experience, and FFM dimensions ($\Delta R^2 = .020$, $p < .01$). No other FFM dimensions significantly predicted this criterion. In step 3, grit significantly predicted bouldering performance: highest redpoint ($\beta = 3.15$, $p < .05$), and highest flash ($\beta = 2.97$, $p < .05$) over age, gender, experience, and FFM dimensions. Overall, grit accounted for 2% of the overall variance in overall bouldering performance over age, gender, experience, and FFM dimensions ($\Delta R^2 = .020$, $p < .01$); for 2.3% in highest redpoint ($\Delta R^2 = .023$, $p < .05$); and for 2.7% in highest flash ($\Delta R^2 = .027$, $p < .05$).

Discussion

This study is the first empirical investigation of the relationship between FFM personality traits, grit, and performance in rock-climbing, within its two separate disciplines, sport climbing and bouldering. Our research expands the already existing efforts of understanding how personality traits are associated with physical activity and athletic performance.

Several key findings emerged: (1) the FFM dimensions predicted different performance criteria in both sport climbing and bouldering, over and beyond age, gender, and experience; (2) Grit consistently predicted different performance criteria in both sport climbing

and bouldering, over and beyond individual FFM dimensions; (3) a post-hoc methodological contribution: although performance in sport climbing and bouldering as measured via the IRCRA is assumed to be multidimensional in nature (including both difficulty rating and style – redpoint, onsight and flash), our analyses suggested that a unidimensional model of climbing performance exhibited the best fit to the data.

#1: FFM personality traits accounted for a significant proportion of variance in rock-climbing performance.

FFM Personality and rock-climbing performance. Overall, our results supported the general hypothesis that personality traits predict rock-climbing performance, in both sport climbing and bouldering. Partly in line with our hypotheses, two traits accounted for most of personality's predictive power in respect to sport climbing and bouldering performance: openness and agreeableness. Although we hypothesised that neuroticism and extraversion would significantly explain variance in the climbing performance outcomes, their impact was negligible.

The criterion validity of FFM traits in explaining performance in sport climbing ranged from 6.5% (highest redpoint) to 7.4% (overall sport climbing – latent). The incremental validity of FFM traits in predicting bouldering performance over age, gender and experience was lower than for sport climbing, but still significant, explaining 5.5% of the variance in bouldering performance.

When zooming in on these findings, openness emerged as having a significant association with all performance measures: overall sport climbing, highest redpoint, highest onsight. In bouldering, openness explained a significant variance in performance. This finding is in line with previous findings outlining openness' importance in respect to high-risk sport participation (e.g., McEwan et al., 2019; Rumbold et al., 2021). Additionally, it expands the current understanding, revealing that openness plays an important role not only in predicting sport participation, but also in accounting for performance variance within this sport. Openness to experience is a personality trait associated with curiosity, creativity, liberal attitudes, and openness to engage in intellectual pursuits. Consequently, we speculate that a potential explanation for its impact over rock-climbing performance resides in the increased likelihood of trying new training methods, new approaches to improve performance-relevant technical or mental skills. At the same time, we emphasise that this finding runs contrary to the common understanding regarding the relatively trivial and negligible role attributed to openness in other domains of human performance, such as work performance (e.g., Schmitt, 2014).

As for the remaining traits for which we expected to encounter significant associations with performance, namely extraversion and neuroticism, our data suggests that their role in climbing performance across the two disciplines is negligible. Extreme sports typically involve exposure to situations that trigger intense affects: heights, exposure, managing life-threatening risks. It may as well be that in order to start practicing such sports in the first place, athletes need a certain degree of emotional robustness, individuals scoring high on neuroticism hardly managing to practice the sport consistently, let alone attain a considerable level of performance over the long run (McEwan et al., 2019). Being a sport with a relatively high level of exposure to dangerous and potentially life-threatening situations, rock-climbing might not be appealing to individuals having

high scores on neuroticism. Therefore, individuals having high levels of neuroticism are likely to select themselves out of practicing rock-climbing. Such personality-based self-selective influences could lead to a restricted range in respect to neuroticism and possible other personality traits, consequently diminishing their observed criterion validity in predicting rock-climbing performance. Consequently, neuroticism is likely to have a less straightforward and linear impact over performance in rock-climbing. In respect to extraversion, its almost zero impact on climbing performance could be explained by the fact that both sport climbing and bouldering are, essentially, individual sports, where performance depends exclusively on the athletes' individual skills and capabilities, rather than on their interpersonal effectiveness.

A surprising and unexpected finding was that agreeableness emerged as a significant predictor of rock-climbing performance across the two disciplines. This finding warrants further explanations. Its negative association with performance in an athletic discipline emerged as significant in other studies. For example, Martin et al. (2011) identified lower levels of agreeableness amongst the athletes qualified for the Paralympic games compared to their peers. There are several ways in which low agreeableness could contribute to attaining higher performance in rock-climbing. First, persons low on agreeableness tend to compete more frequently with others. Even though rock-climbing is an individual sport, practitioners often get ranked based on their individual performance and highest grades (redpoints or onsight) are regarded as a proxy for the practitioners' skills. Second, low agreeableness might enable some practitioners to be more aggressive when performing the sport and/or during training sessions. Third, agreeableness emerged as a significant predictor only for outdoor sport climbing, but not in bouldering, which could be explained by the more solitary nature of the outdoor sport climbing, as opposed to the more group-oriented specificity of bouldering.

Several broader conclusions emerge from our findings regarding personality and rock-climbing performance. First, our data suggest that personality traits are differentially related with sport participation and sport performance, respectively. Consequently, caution should be used when generalising the results of previous research where mere participation and not performance was measured. Having an interest in a certain sport or having the "personality profile" specific to some sport, such as high-risk sports does inform about the likelihood of attaining success or a high performance in the respective discipline. Second, contrary to the expectation articulated by Allen et al. (2013, p. 187) that personality will predict "... athletic success based on the theoretical predictions that govern associations between personality and academic or organizational success," our findings revealed that personality traits have a rather negligible impact over work performance, such as openness, coherently predict performance in rock-climbing. Conversely, other traits that have been traditionally associated with work performance, such as conscientiousness (e.g., Schmitt, 2014) are relatively irrelevant in explaining performance in rock-climbing. Third, our findings contribute to the current conversation regarding the relevance of personality traits for athletic performance. In line with the arguments made by Gill and Williams (2008), despite its significant association with rock-climbing performance, personality accounted for a mere 5% to 7% of the performance variance in rock-climbing, even though others regard it as being an important antecedent of athletic performance (Möttus et al., 2017; Newland et al., 2020; Wilson & Dishman, 2015). In contrast, experience (years of practice) accounted for approximately 20% of the variance

in rock-climbing performance, suggesting that other factors acquired via experience, such as technical skills or sport-specific physiological adaptations might play a critical role in explaining performance within this discipline.

#2: Grit explained a significant amount of variance in rock-climbing performance over age, gender, experience, and FFM personality traits.

Grit alone explained significant amounts of personality-related variance in rock-climbing performance, across both disciplines – sport climbing and bouldering, respectively. Grit's criterion validity in explaining performance in sport climbing ranged from 2.7% (overall sport climbing – latent) to 3.1% (highest redpoint). Its predictive validity in respect to bouldering was lower than for sport climbing, but still significant, ranging from 2% (overall bouldering performance – latent) to 2.7% (highest flash) explained variance.

With respect to explaining the outcomes of focus, as a standalone trait, grit has a higher explanatory value compared to the dimensions of the more popular and well-established FFM model in accounting for variance in athletic performance – Credé et al. (2017) reported similar findings in respect to academic performance. Single-handedly, grit predicted climbing performance over age, gender, experience, better than any other single FFM trait in respect to all the five performance criteria and across the two disciplines (sport climbing and bouldering). This finding is unsurprising in the light of previous investigations. For example, Harper et al. (2018) reported that grit was associated with the ability to exert significantly more physical effort. Moreover, several investigations support the conclusion that grittier individuals are more likely to spend time in domain-specific activities and end up accumulating longer periods of practice, which in turn facilitates higher levels of performance and discipline-specific attainment (Ford & Williams, 2012). Moreover, grittier individuals are more likely to accumulate physical and non-physical sport-specific activities (Larkin et al., 2016). Investigating the association between this personality construct and various aspects of athletic performance and practice, Tedesqui and Young (2018) observed that athletes with higher persistence (a facet of grit) reported higher volumes of deliberate practice. A relatively recent empirical investigation observed that elite athletes reported higher scores on Grit compared to a large sample of non-athletes (From et al., 2020).

#3: A one factor model of rock-climbing performance exhibited a good fit to the data collected with the IRCRA scale

Performance in rock climbing is typically assessed via several key indicators, such as Highest Redpoint (routes where several attempts are needed free climb them), Onsight, Flash. The rock-climbing community, as well as the official IRCRA scale performance in rock-climbing typically distinguishes between these five performance categories. For example, the popular rock-climbing portal (www.8a.nu) uses the same types of entries. These three performance criteria are regarded as requiring different physical and mental skills. Our investigation reveals that irrespective of these conceptual distinctions, the three criteria actually load onto a single global rock-climbing performance factor, which suggests that a unitary measure might best capture the variability of performance in this sport, as opposed to the largely used five categories.

Limitations

There are several issues that would warrant further discussion. First, our investigation relied on a cross-sectional design, where antecedents and outcomes are measured at the same point in time. This procedural design does not permit us to establish the direction of the influence. Reverse causal effects, such as personality changes caused by regularly practicing rock-climbing are viable hypotheses that cannot be tested in this design. In this sense, further directions of research would include longitudinal designs and the inclusion of other possible outcomes, especially since the author of the scale has claimed that this personality construct can change over the course of a person's lifetime, as a result of effort, environment and others (Duckworth, 2016).

Second, the results could be plagued by common method bias, as both predictor- and criterion-related measures were collected simultaneously. Finally, other specific personality dimensions which were not included (e.g., impulsivity or sensation-seeking) could add to the multifaceted nature of the interrelationships between personality and performance in this high-risk sport.

Third, we drew on a sample of international rock-climbing amateur practitioners and, consequently, we could not account for the effects of potential cross-cultural differences or those differences between professional and amateur practitioners.

Conclusions

Understanding the way in which personality constructs, such as the FFM and grit, can contribute to explaining the variance of climbing performance outcomes is an important steppingstone in establishing the practical values of these constructs. Taken together, our findings suggest that grit, openness, and agreeableness had a relevant impact over rock-climbing performance.

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Data availability statement

The data that support the findings of this study are available from the corresponding author, AI, upon reasonable request.

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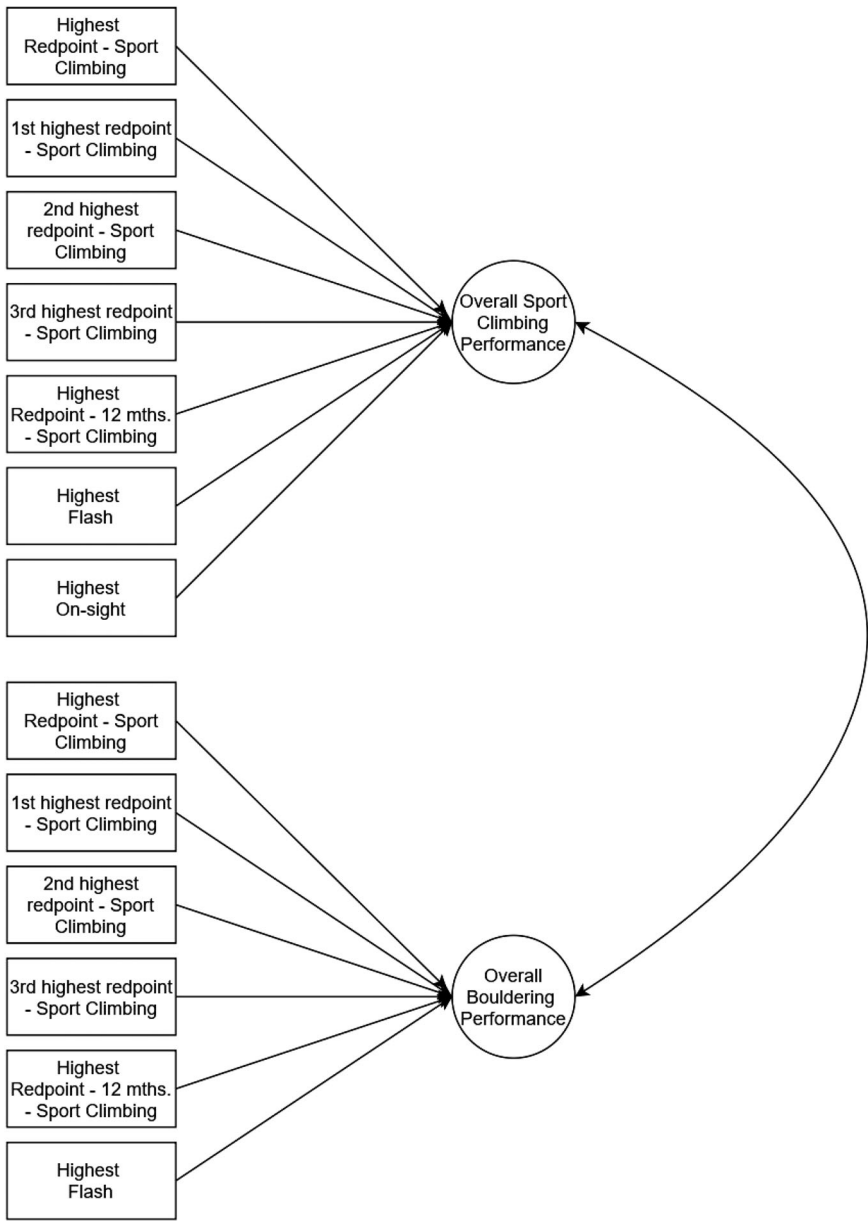
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Appendices

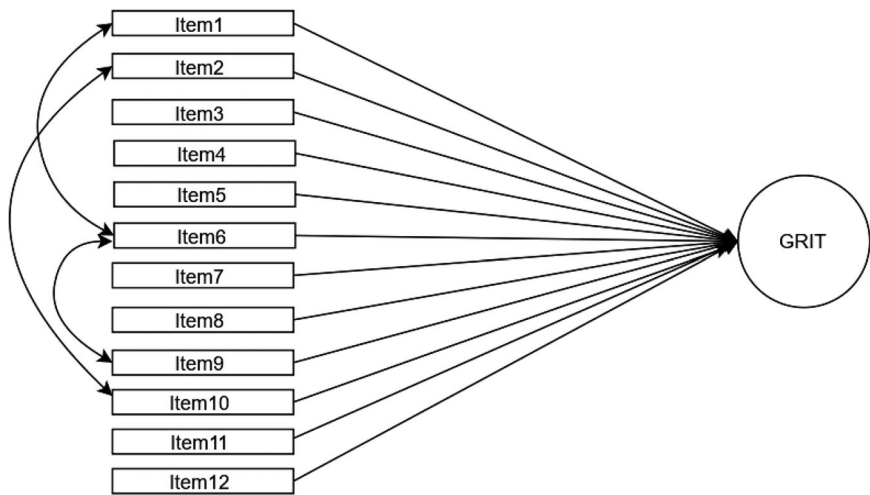
Appendix 1. Reporting grades in climbing research

Climbing Group	Vermint Font		IRCRA		YDS		French/sport		British Tech		Ewbank		BRZ		UIAA		Metric	
	Reporting Scale		Reporting Scale		Reporting Scale		Reporting Scale		Reporting Scale		Reporting Scale		Reporting Scale		Reporting Scale		Reporting Scale	
Lower Grade (Level 1) Male & Female			1		5.1		1				2		I sup		I		1.00	
			2		5.2		2				6		II		II		2.00	
			3		5.3		2+				8		II sup		III		3.00	
			4		5.4		3-				10		III		III+		3.50	
			5		5.5		3				12		IV		IV		4.00	
			6		5.6		3+				14		V		V-		4.33	
			7		5.7		4				16		V sup		V		4.66	0.00
			8		5.8		4+				18		V+		V+		5.00	0.25
Intermediate (Level 2) Male	VB	< 2	9		5.9		5				20		VI		VI-		5.33	0.50
			10		5.10a		5+				22		VI sup		VI		5.66	0.75
			11		5.10b		6a				24		VI		VI+		6.00	1.00
			12		5.10c		6a+				26		VI+		VI+		6.33	1.25
			13		5.10d		6b				28		VI sup		VII-		6.66	1.50
			14		5.11a		6b+				30		VII		VII		7.00	1.75
			15		5.11b		6c				32		7a		VII+		7.33	2.00
			16		5.11c		6c+				34		7b		VIII-		7.66	2.25
Advanced (Level 3) Female			17		5.11d		7a				36		7c		VIII		8.00	2.75
			18		5.12a		7a+				38		8a		VIII+		8.33	3.00
			19		5.12b		7b				40		8b		IX-		8.66	3.25
			20		5.12c		7b+				42		8c		IX		8.99	3.50
			21		5.12d		7c				44		9a		IX		9.33	3.75
			22		5.13a		7c+				46		9b		IX+		9.66	4.00
			23		5.13b		8a				48		9c		X-		9.99	4.25
			24		5.13c		8a+				50		10a		X		10.33	4.50
Elite (Level 4) Male			25		5.13d		8b				52		10b		X		10.66	4.75
			26		5.14a		8b+				54		10c		X+		11.00	5.00
			27		5.14b		8c				56		11a		XI-		11.33	5.25
			28		5.14c		8c+				58		11b		XI		11.66	5.50
			29		5.14d		9a				60		11c		XI+		11.99	5.75
			30		5.15a		9a+				62		12a		XII-		12.33	6.00
			31		5.15b		9b				64		12b		XII		12.66	6.25
			32		5.15c		9b+				66		12c		XII		13.00	6.50

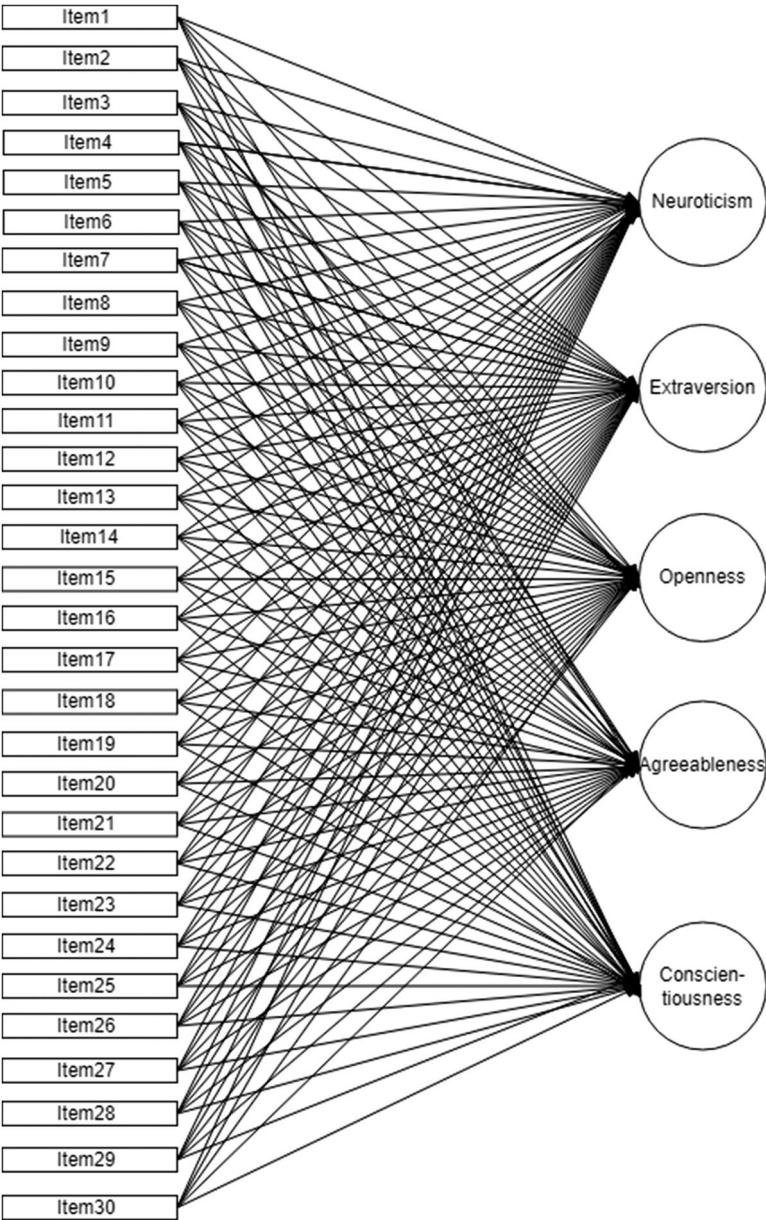
Appendix 2. CFA model for IRCRA scale



Appendix 3. CFA model for Grit Scale



Appendix 4. ESEM model for Big Five Inventory–2 Short Form



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